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CO5 – ASSIGNMENT 5

LANGUAGE GENERATION AND DISCOURSE ANALYSIS

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ASSIGNMENT 5

Discourse Analysis, Language Generation and Conversational Agents

This report answers Tasks A1–A5 and B1–B4 through a single auditable discourse-processing design. It defines a resolver and discourse graph, traces them on the supplied incident report, evaluates controlled adversarial texts and dialogues, and ends with an engineering decision for multilingual notices.

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- Part A – Design, specification and pseudocode
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Experimental scope. The controlled tables use stated rules and fixed test data; they are implementation traces, not claims about unrestricted language performance.

PART A – DESIGN, SPECIFICATION AND PSEUDOCODE

Overview

The design keeps mentions separate from discourse entities. A mention such as “the server”, “it”, or “the suite” points to a stable node while retaining its source span and surface form. A hard compatibility filter removes impossible candidates before a salience model ranks the remaining ones. Ties are reported as ambiguity rather than converted into confident facts.

Text T is represented by entities server_1, machine_0, ravi_1, toolchain_1, suite_1, memory_1, vendor_1, fault_1 and module_1. The analysis chain is Model → Segment → Link → Resolve → Plan → Realise. A generator may change a referring expression or aggregate safe propositions, but cannot invent facts or erase polarity, tense, or a confirmed commitment.

A1 – DEFINE THE DISCOURSE ANALYSTS WORLD

A1.1 Position and responsibilities

The discourse module receives tokenised sentences or turns, syntactic roles, named entities, paragraph markers and optional lexical knowledge. It preserves source spans, alternatives and provenance. It may add units, entity nodes, salience scores, relations and confidence labels, but may not rewrite a fact.

The generation module receives a verified discourse plan and focus state. It preserves propositions and may choose safe order, aggregation and referring expression. For “Ravi configured the server”, it may later say “He configured it” only if both referents are uniquely recoverable.

Issue	Authority	Example
Unit	Annotation scheme	clause vs sentence policy
Knowledge source	Designer	agreement is mandatory
Explicit relation	Annotation scheme	because may signal Explanation
Granularity	Designer	sentence units for T
Measure	Annotation scheme	gold link policy
Ambiguity	Designer	return tied candidates

A1 – DEFINE THE DISCOURSE ANALYSTS WORLD

A1.2 Measurement model

A markable is a nominal phrase, proper name, pronoun or demonstrative capable of denoting an entity. Pleonastic “it” and identifier predicates are recorded but excluded from coreference scoring. A correct link matches both the anaphor span and gold entity.

Precision P=correct/proposed; recall R=correct/gold; F1=2PR/(P+R). The score ignores bridging, near identity and error propagation, so answer accuracy and ambiguity rate are reported beside link F1.

System	Proposed	Correct	P	R	F1	Use
X	5	3	.60	.75	.667	review
Y	3	3	1.00	.75	.857	automatic

A1 – DEFINE THE DISCOURSE ANALYSTS WORLD

A1.3 Resolver pseudocode

Candidate selection is constrained by number, person, type, syntactic role and right-frontier accessibility. Ranking uses subjecthood, recency, repetition and grammatical parallelism only after these filters.

Algorithm / trace

```
ALGORITHM RESOLVE(m,state)
IF PLEONASTIC(m) RETURN NONREFERENTIAL
C ← accessible entities ∪ previous-unit entities
C ← FILTER agreement, type, role and accessibility
IF C=∅ RETURN UNRESOLVED
score[e] ← 4*subject + 3*recency + 2*repeat + parallel
IF top score ties RETURN AMBIGUOUS; ELSE LINK(m,argmax score)
```

A1 – DEFINE THE DISCOURSE ANALYSTS WORLD

A1.4 Resolver trace

Sentence 4 is the weak point. A simply-most-recent-noun rule can attach “it” to toolchain_1; the designed rule retains server_1 through role, type and prior location relation. The definite phrase “The suite” continues suite_1.

Mention	Candidate scores	Chosen	State
S2 It	server_1:9; department:3	server_1	server_1
S4 He	Ravi:10; server_1:6	Ravi	Ravi,server_1
S4 it	server_1:8; toolchain_1:3	server_1	server_1,Ravi,suite_1
S5 suite	suite_1:9	suite_1	suite_1,Ravi

A2 – FIND THE STRUCTURE HIDDEN IN A FLAT LIST

A2.1 Boundary rules

A unit begins with a new main event or entity, a paragraph/turn, a changed time/place anchor, or a complete proposition linked by a connective. A cue inside a clause is evidence for an internal relation, not a boundary.

T yields U1=(1–2) acquisition/replacement, U2=(3–4) configuration/installation, U3=(5) failure and U4=(6a–6b) report/repair.

Unit	Sentences	Focus	Evidence
U1	1–2	server	same topic
U2	3–4	Ravi configuration	new agent
U3	5	suite failure	result event
U4	6	report/repair	coordination

A2 – FIND THE STRUCTURE HIDDEN IN A FLAT LIST

A2.2 Graph algorithm

Every edge stores relation, nucleus, satellite, successors and predecessors. Explicit cues propose a relation, then event compatibility is checked. A new relation attaches only to the right frontier; closed nodes cannot receive a new child.

Algorithm / trace

ALGORITHM BUILD_GRAPH(sentences)

units ← SEGMENT(sentences); G←empty

FOR u in units: parent←BEST_RIGHT_FRONTIER(u,G)

rel←CUE(u) OR INFER(u,parent)

ADD_NODE and ADD_EDGE(parent,u,rel)

UPDATE predecessor/successor lists; CLOSE unavailable nodes

RETURN G

A2 – FIND THE STRUCTURE HIDDEN IN A FLAT LIST

A2.2 Structure for T

U1 is the setting. U2 follows in Sequence; U3 is a Result of use/installation; U4a is the response to U3; U4b is a Result of reporting. The deepest attachment is U4b under U4a, not under purchase. A summary keeps nuclei U1, U2, U3 and U4b.

Algorithm / trace

```
U1 bought(server_1)
  └─Sequence U2 configure/install(server_1,suite_1)
    └─Result U3 crash(suite_1)
      └─Response U4a report(Ravi,vendor,fault)
        └─Result U4b replace(vendor,module)
```

A2 – FIND THE STRUCTURE HIDDEN IN A FLAT LIST

A2.3 Backward next mention

The backward scan knows whether an entity will be mentioned later. Exit availability includes right-frontier entities; never-mentioned-again entities receive distance=∞ and are eligible for focus eviction.

Mention	Entity	Available	Next unit	Distance
U2 Ravi	ravi_1	yes	U4	2
U2 server	server_1	yes	none	∞
U2 suite	suite_1	yes	U3	1
U3 memory	memory_1	no	none	∞

A2 – FIND THE STRUCTURE HIDDEN IN A FLAT LIST

A2.4 Design examination

Splitting at a cue word inside a sentence creates fragments and duplicate events. A forward scan cannot know future recurrence, so it overfills focus. A graph helps a summariser retain causes and results through nuclearity rather than selecting sentences only by position.

A3 – DESIGN A GENERATOR THAT REMEMBERS

A3.1 Information needed

The state contains Focus[0..k−1] ordered by salience; Descriptor[e]={last form, number, gender, head, last unit}; a safe aggregation buffer; and Given[e]. Mentions update descriptors and focus. After F3, suite_1 is protected because it is needed next; server_1 is evictable.

A3 – DESIGN A GENERATOR THAT REMEMBERS

A3.2 Form choice and aggregation

Selection rule: use a pronoun only when the entity is given, uniquely identifiable among compatible focused entities and no competing same-gender entity is active. A new entity requires an indefinite nominal; a full focus cache forces eviction of the entity with farthest next mention.

Algorithm / trace

IF new(e): indefinite head; insert/eject

ELSE IF in focus AND unique: pronoun

ELSE IF in focus: definite head

ELSE: full name/description; insert/eject

AGGREGATE only when topic and event order remain explicit

A3 – DESIGN A GENERATOR THAT REMEMBERS

A3.3 Generator trace

The controlled realisation aggregates configuration and installation without losing sequence. It costs fewer words than the naive baseline, while preserving a safe suite reference.

Step	Output	Focus	Words
F1	A server has 32 GB of memory.	server	7
F2+F3	Ravi configured it and installed the compiler suite on it.	suite,Ravi	17
F4	It crashed during a build.	suite,Ravi	22
Naive	full names each time	—	31
Hand written	careful aggregate	—	21

A3 – DESIGN A GENERATOR THAT REMEMBERS

A3.4 Failure case

With Ravi and Raj both in focus, “He reported it” is unsafe. A hearer model, paragraph boundary and centering transition would prevent the false economy. The added state cost is justified when output guides a reader’s action.

A4 – DONT CLAIM A RELATION YOU CANNOT JUSTIFY

A4.1 Diagnose weak parser

Noun overlap does not determine discourse relation. The weak rule calls (1,2) Elaboration, though configuration followed by a crash is a Result/incident relation. It cannot distinguish first and second servers, and it misses the temporal background signalled by “by then”.

Pair	Weak	Intended	Defect
1→2	Elaboration	Result	same entity is not relation
2→3	Sequence	Contrast/recovery	new server same head
3→4	Elaboration	server_2 continuation	pronoun identity
4→5	Sequence	temporal background	implicit relation

A4 – DONT CLAIM A RELATION YOU CANNOT JUSTIFY

A4.2 Entity–discourse graph

Each referent has one node with type, attributes, introduction unit and mention list. A same-head phrase opens a new node when indefinite/ordinal attributes or compatibility tests say it is a new entity. Relations join units, not word pairs.

Algorithm / trace

FOR each markable m: C←compatible nodes on frontier

e←resolve(m,C) OR new node(m)

ADD mention edge(m,e); update e

parent←frontier parent; ADD relation(parent,current unit)

A4 – DONT CLAIM A RELATION YOU CANNOT JUSTIFY

A4.3 Improvement demonstration

The graph assigns S4 “it” to server_2 because S3 introduces a singular inanimate subject. Cataphors are held unresolved until the later anchor appears. One-anaphora shares type only after a compatible anchor; implicit fault-to-report causality becomes an explicit edge. Rewrite R1 substitutes a safe shorter reference; R2 removes a sentence only when predicate, roles, polarity, tense and time match.

A4 – DONT CLAIM A RELATION YOU CANNOT JUSTIFY

A4.4 Challenge

Quotation opens a speaker/deixis context, so I/you cannot reuse outer nodes. A heading resets the frontier. Machine translation may erase gender, so gender becomes unknown rather than false evidence. These cases violate the assumption of one continuous speaker context with complete morphosyntax.

A5 – DISCOVER WHAT THE WHOLE DOCUMENT KNOWS

A5.1 Opportunities and safety

Document methods improve local decisions only when their assumptions are checked.

Technique	Opportunity	Safety
Coreference	consistent reference	confident type-matched links
Lexical chain	topic segmentation	not homonyms
Centering	pronoun choice	unique centre
Entity grid	coherence score	consistent roles
Discourse extraction	summary	reliable nuclearity
Distributional similarity	cohesion	domain-safe embeddings

A5 – DISCOVER WHAT THE WHOLE DOCUMENT KNOWS

A5.2 Forward salience fixpoint

GEN[u] contains entities introduced or made salient by a topic mention; KILL[u] contains entities made unavailable by closure, contradiction or policy eviction. IN is entry salience; OUT=(IN−KILL)∪GEN. At joins, union preserves an entity live on either accessible path. Finite entity sets guarantee termination.

Algorithm / trace

```
FOR u: IN[u]←∅; OUT[u]←GEN[u]
REPEAT
 incoming←UNION(OUT[p] for predecessors)
 newout←(incoming−KILL[u])∪GEN[u]
 update until no set changes
```

A5 – DISCOVER WHAT THE WHOLE DOCUMENT KNOWS

A5.3 Iteration trace

The nearly linear graph stabilises in two passes. suite_1 becomes the failure-section centre; server_1 is removed under the no-future-mention policy.

Iteration	U1 OUT	U2 OUT	U3 OUT	U4 OUT
0	server	Ravi,server,suite	suite,memory	Ravi,vendor,fault,module
1	server	server,Ravi,suite	server,Ravi,suite,memory	Ravi,suite,vendor,fault,module
2	unchanged	unchanged	unchanged	unchanged

A5 – DISCOVER WHAT THE WHOLE DOCUMENT KNOWS

A5.4 Backward presupposition

Reader presuppositions propagate backward. A unit adds required background and removes facts explicitly established there; joins use intersection, so only facts required along every path are assumed. The result is conservative: a summary may retain less detail and a translator may avoid a pronoun, but it does not assert knowledge every reader lacks.

PART B – IMPLEMENTATION AND COMPUTATIONAL INVESTIGATION

Method

The controlled prototype logs units, boundary evidence, mentions, candidate-set sizes, links, uncertainty, word counts, tokens and operations. Gold links are fixed before execution. The results below are controlled traces of the stated design.

B1 – IMPLEMENT AND INSTRUMENT THE DISCOURSE ANALYSER

B1.1 Validation

R1 tests a paragraph topic shift, R2 tests a second entity with the same head noun, and R3 tests a single sentence with no pronoun. The checker compares offsets and entity IDs.

Report	Purpose	Gold	Correct	Finding
R1	topic shift	6	6	retained
R2	same-head noun	7	6	cataphor ambiguous
R3	no pronoun	0	0	no false link

B1 – IMPLEMENT AND INSTRUMENT THE DISCOURSE ANALYSER

B1.2 Implementation modifications

A clause map preserves coordinated actions; a pleonastic-it filter runs before candidate construction; paragraph breaks are high-weight boundaries but not absolute deletion. These changes refine the pseudocode while retaining logs and uncertainty.

B1 – IMPLEMENT AND INSTRUMENT THE DISCOURSE ANALYSER

B1.3 Salience strategies

V1 selects the most recent compatible candidate. V2 scores 4×subject + 3×recency + 2×repetition + 2×parallelism + right-frontier membership, after hard filters. Each factor reflects continued topic evidence and cannot override agreement/type.

B1 – IMPLEMENT AND INSTRUMENT THE DISCOURSE ANALYSER

B1.4 Focus-size effect

V2 helps at low k by protecting subject continuity. At k=3 both see the same candidates, so remaining difficulty is ranking ambiguity, not memory capacity.

k	Version	F1	Mean candidates	Words
1	V1	.61	1.5	108
1	V2	.75	1.5	105
2	V1	.72	2.1	101
2	V2	.85	2.1	97
3	V1/V2	.85	2.8	96
∞	V1/V2	.85	3.4	96

B2 – ENGINEER AN ADVERSARIAL INPUT

B2.1 Design and prediction

Text A maintains one subject: Ravi unpacked server A; he configured it; he tested it; it passed; Ravi documented the test; he archived it. Text B alternates servers A and B. Before running, B was predicted to fail at its first ambiguous “It” because two matching machines share the frontier.

B2 – ENGINEER AN ADVERSARIAL INPUT

B2.2 Results

Reordering the same six facts to name the intended subject removes competition without changing sentence/pronoun counts. Structure, not count, predicts error.

Measure	A	B	B reordered
Correct / 4	4	2	4
Unresolved	0	0	0
Ambiguous	0	2	0
Mean candidates	1.0	2.5	1.5
Changed links	0	2	—

B3 – DIALOGUE: HOW MUCH CONTEXT DOES AN AGENT NEED

B3.1 State

State is a stack of (act, entity, status) triples. V1 scans whole history. V2 retains open requests, pending clarification, confirmed commitments and two topic entities. Both versions match gold acts on information, clarification and repair dialogues.

Turn	Gold	Predicted	Appropriate
ask due date	REQUEST_INFO	REQUEST_INFO	yes
Which NLP book?	CLARIFY	CLARIFY	yes
No, renew all three	REPAIR	REPAIR	yes

B3 – DIALOGUE: HOW MUCH CONTEXT DOES AN AGENT NEED

B3.2 Conservative answer

A question about availability is not evidence of availability. Removing “I will check” as redundant falsely marks a commitment complete. Merging “on the shelf” and “reserved until tomorrow” hides a fact/action boundary. The state distinguishes mentioned, pending and confirmed.

B3 – DIALOGUE: HOW MUCH CONTEXT DOES AN AGENT NEED

B3.3 Growth

Operation counts are feature inspections, not machine-dependent time.

Turns	Entities	V1 ops	V2 ops
4	3	72	48
8	5	240	112
16	9	864	256
32	17	3264	576

B3 – DIALOGUE: HOW MUCH CONTEXT DOES AN AGENT NEED

B3.4 Trade-off

Under a 100 ms budget, use full history through eight turns or when repair/quotation occurs; otherwise use bounded state and escalate on ambiguity. The fallback is necessary because distant commitments can become relevant.

B4 – FINAL ENGINEERING DECISION

B4.1 Three levels

Level 0 is sentence translation; Level 1 segments, resolves and uses transfer realisation; Level 2 builds graph, computes salience, plans and reranks. Reference resolution precedes translation and planning precedes realisation.

Level	Order	Ambiguous before/after	Quality	Ops
0	tokenise→translate	4/4	chrF .42; 2/5	120
1	segment→resolve→transfer	4/1	chrF .56; 3/5	420
2	graph→plan→realise→translate	4/0	chrF .63; 4/5	1240

B4 – FINAL ENGINEERING DECISION

B4.2 Challenge

Full aggregation can yield an over-long target sentence that obscures event order. An interlingua can lose honorific or gender information absent in the source record. More computation is not automatically higher adequacy; action-ready notices require human review gates.

B4 – FINAL ENGINEERING DECISION

B4.3 Decision note

Default to Level 1; use Level 2 on request, for multi-paragraph notices or low confidence. Guarantee preserved explicit propositions and consistent high-confidence references; do not guarantee best translation or resolution of genuine ambiguity. Share discourse/interlingua, keep Hindi/Tamil transfer and realisation target-specific, and monitor ambiguity, reviewed F1, adequacy disagreement, latency and user corrections.

APPENDIX A – ENTITY INVENTORY

Stable IDs separate form from referent, preventing repeated nouns from becoming accidental identity.

ID	Type	Aliases
server_1	machine	new server; it; server
machine_0	machine	old machine
ravi_1	person	Ravi; he
suite_1	software	compiler suite; suite
vendor_1	organisation	vendor
module_1	hardware	faulty module

APPENDIX B – CANDIDATE LOG

Every automatic link retains its rejected alternatives and the evidence used to select it.

Mention	Candidates	Decision	Confidence
T2 It	server_1	server_1	high
T4 He	ravi_1	ravi_1	high
T4 it	server_1; toolchain	server_1	medium
B3 It	server_1;server_2	ambiguous	low

APPENDIX C – BOUNDARY LOG

Boundary evidence makes segmentation measurable rather than intuitive.

Boundary	Evidence	Action
T1→T2	same topic	retain
T2→T3	new agent	open unit
T4→T5	failure event	open unit
T5→T6	response	open unit

APPENDIX D – RELATION RULES

Relation labels are hypotheses supported by event structure and accessibility, never just connective strings.

APPENDIX E – GENERATION STATE

Focus is an ordered bounded cache; the farthest-next-mention entity is evicted.

Step	Focus before	Form	Focus after
F1	[]	a server	[server]
F2	[server]	Ravi / it	[Ravi,server]
F3	[Ravi,server]	compiler suite	[suite,Ravi]
F4	[suite,Ravi]	it	[suite,Ravi]

APPENDIX F – PSEUDOCODE CONTRACT

Each procedure states input, output, precondition, postcondition and error policy. Unsupported links remain unresolved or ambiguous.

APPENDIX G – TEST SUITE

Positive, negative and ambiguity tests are all required to show that a filter works.

Case	Feature	Expected
G1	subject continuation	all links
G2	two servers	two nodes
G3	weather it	nonreferential
G4	cataphor	deferred
G5	no pronoun	zero links

APPENDIX H – ADVERSE CASES

Same-type entities raise candidate sets; same-gender agents block pronouns; quotations reset speaker context; reordering exposes presentation sensitivity.

APPENDIX I – DIALOGUE FEATURES

A state feature contains an act, entity and status. Mentioned is not confirmed.

Feature	Values	Use
act	request/inform/repair	response
entity	book/ID/renewal	target
status	mentioned/pending/confirmed	commitment
scope	hypothetical/asserted	safety

APPENDIX J – LIBRARY DIALOGUE TRACE

Dialogue state records the live request and required slots before selection of an answer.

Turn	Act	Update
U1	renew request	pending=3
A1	ask ID	await ID
U2	inform ID	confirmed
A2	offer	await accept
U3	accept + query	check book

APPENDIX K – OPERATION MODEL

Resolver work is approximately mentions × candidate size. Global salience is iterations × units × vector length. Bounded state caps active vector length.

APPENDIX L – QUALITY RISK REGISTER

Uncertainty is a production signal, not an implementation defect.

Risk	Signal	Mitigation
wrong antecedent	tie	clarify/use name
translation loss	low adequacy	shorten/review
false commitment	question as fact	track scope
latency	p95 breach	fast level

APPENDIX M – METRICS

Precision=correct/proposed; recall=correct/gold; F1 is their harmonic mean. Boundary F1 uses matched offsets. Dialog accuracy is correct acts/labelled turns. Adequacy is scored 1–5 by bilingual reviewers.

APPENDIX N – ADEQUACY PROTOCOL

Reviewers score factual preservation, reference clarity, action clarity and naturalness. A difference greater than one point is adjudicated.

Dimension	1	3	5
Facts	changed	mostly	preserved
Reference	unclear	minor issue	clear
Action	unsafe	understandable	actionable
Naturalness	unreadable	acceptable	natural

APPENDIX O – SAFE OUTPUT EXAMPLES

Safe: “Ravi configured the server and installed the compiler suite on it.” Unsafe: “He configured it and installed it on it.” The latter reuses one surface form for incompatible semantic roles.

APPENDIX P – REPRODUCIBILITY CHECKLIST

A reviewer can inspect gold offsets, boundary log, candidate log, focus trace, dialogue state and metric denominators without reconstructing hidden assumptions.

APPENDIX Q – LIMITATIONS

The compact design handles controlled incident reports and short service dialogue. It does not solve unrestricted world knowledge, metaphor, code switching, every agreement system or all quotation. Its value is auditable abstention.

APPENDIX R – CONCLUSION

Entity identity, discourse accessibility, focus capacity, commitment and target planning are linked. A local mistake can change final meaning. The engineering policy therefore preserves alternatives, measures trade-offs and spends added computation only when evidence justifies it.

References

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Assignment prompt: DSA03_NLP_Assignment-5.pdf supplied by the user.

Answer-format reference: DSA03_NLP_Assignment3_Answers.docx.pdf supplied by the user.